

Samuel Hurh

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Software Engineer — XR Systems, Wearable AI & Real-Time 3D

EDUCATION

University of Illinois at Urbana-Champaign

Bachelor of Science in Computer Science

Champaign, IL

Aug. 2020 – Aug. 2024

University of Illinois at Urbana-Champaign

Master of Computer Science (MCS)

Champaign, IL

Jan. 2025 – May 2026

TECHNICAL SKILLS

Languages: C#, Python, C++, C, CUDA, JavaScript, Kotlin

XR & Real-Time 3D: Unity, XRI, Meta SDK, Unity Netcode, OpenXR, Unreal Engine 5, Magic Leap 1/2, Meta Quest 2/3, Google Glass EE2, WebGL

Perception & AI: RealSense D455, RGB-D sensing, Open3D, ICP, 6-DoF pose estimation, computer vision pipelines, eye tracking, PyTorch, Gemini / Vertex AI, speech-to-text

Systems: Android, Cloud Run, real-time pipelines, state machines, networking (TCP/UDP), RTC/media workflows, MPI, CUDA

EXPERIENCE

XR Software Development Contractor Intern

Kohler

March 2024 – June 2026

Champaign, IL

- Designed modular AR/CV workflows aligning sensed physical-world geometry with CAD context using RGB-D data, point-cloud processing, and ICP-based geometric registration.
- Evaluated model-based 6-DoF pose estimation approaches such as FoundationPose against classical registration methods under dataset, compute, and robustness constraints.
- Built Unity/Magic Leap visualization prototypes for spatial review, calibration, and high-level manufacturing deviation visualization.
- Architected an internal multi-user VR collaboration prototype using Unity networking and RTC-style media workflows to replace an external vendor solution.
- Prototyped a UE5 room-scale VR product visualization experience for stakeholder review of product scale and spatial layout.

VR Research Engineer

The Studio at UIUC

Sep. 2025 – April 2026

Champaign, IL

- Built the core interaction architecture for a VR professional training simulator, integrating natural-language input into a stateful real-time system.
- Designed state-machine-driven dialogue control that constrained model outputs to predefined semantic transitions for predictable immersive interaction.

Undergraduate Research Assistant

University of Illinois at Urbana-Champaign

Sep. 2023 – May 2025

Champaign, IL

- Implemented gaze-depth interaction prototypes on Magic Leap using eye-tracking convergence signals for hands-free AR UI actuation.
- Prototyped gaze-swipe typing with autocomplete support and time-step gaze clustering to explore dwell-free XR text input.
- Built demo and evaluation materials for AR/VR HCI research on gaze-driven interaction techniques.

PROJECTS

Google Glass AI Assistant | *Android/Kotlin, Cloud Run, Gemini / Vertex AI*

June 2026 – Present

- Built a wearable AI assistant prototype for Google Glass EE2 using speech input, image capture, and constrained heads-up display output.
- Connected Android client flow to Cloud Run services for speech-to-text, Gemini / Vertex AI, and image-aware responses.

Unity VR Interaction & Roguelike Systems | *Unity, C#, Meta Quest*

May 2023 – May 2026

- Expanded an early VR interaction sandbox into a playable roguelike prototype with procedural layouts, weapon systems, and enemy behavior.
- Built interaction-heavy systems around grabbing, recoil, targeting, combat flow, procedural room placement, and replayable spatial layouts.

VR High-Frequency Trading Data Visualization | *Unity, C#, Meta Quest*

Sep. 2024 – Dec. 2024

- Engineered a VR data visualization prototype for historical and live order book data, market depth, and exchange packet-flow concepts.
- Modeled inter-exchange packet transfer behavior using structured event simulation based on real packet information.